

TABLE 22 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.88	4.56	5.36	6.30	7.41	8.74	10.33	12.26	18.0%
Japan	1.80	2.14	2.52	2.98	3.53	4.18	4.97	5.93	18.6%
India	0.23	0.27	0.32	0.37	0.44	0.51	0.61	0.72	17.8%
Rest of Asia Pacific	3.25	3.82	4.48	5.25	6.16	7.24	8.54	10.10	17.7%
Total	9.17	10.79	12.68	14.91	17.53	20.67	24.44	29.01	18.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 23 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SEQUENCING AND LIBRARY PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.98	1.10	1.22	1.36	1.52	1.70	1.91	2.14	11.9%
Rest of Latin America	1.11	1.24	1.38	1.53	1.71	1.91	2.14	2.40	11.8%
Total	2.09	2.33	2.60	2.90	3.23	3.61	4.04	4.54	11.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.4 DATA ANALYSIS

6.4.1 DEVELOPMENT OF ADVANCED BIOINFORMATICS SOLUTIONS TO DRIVE GROWTH

This segment includes bioinformatics/data analysis services for samples subjected to microbiome sequencing. Researchers in the genomics field often outsource the data analysis part of their projects to bioinformatics service providers. NGS data analysis services are offered for data processing and quality control, assembly, mapping, annotation, and advanced analysis for microbiome research and population studies.

The rising awareness about NGS data analysis services, the recent launch of data analysis services by bioinformatics companies, and the increasing availability of data analysis services in developed countries are the major factors supporting the growth of this market.

TABLE 24 MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	18.95	21.38	24.11	27.21	30.73	34.83	39.61	45.25	13.4%
Europe	10.37	11.72	13.24	14.97	16.93	19.22	21.89	25.05	13.6%
Asia Pacific	2.89	3.33	3.83	4.40	5.06	5.84	6.75	7.83	15.4%
Latin America	0.66	0.72	0.78	0.85	0.92	1.00	1.09	1.19	8.7%
Middle East and Africa	0.46	0.49	0.53	0.56	0.60	0.64	0.68	0.73	6.8%
Total	33.34	37.64	42.48	47.99	54.24	61.53	70.03	80.04	13.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 25 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	17.17	19.39	21.88	24.72	27.94	31.70	36.09	41.26	13.5%
Canada	1.78	1.99	2.22	2.49	2.78	3.13	3.53	3.99	12.4%
Total	18.95	21.38	24.11	27.21	30.73	34.83	39.61	45.25	13.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 26 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR DATA ANALYSIS, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	2.23	2.54	2.89	3.28	3.74	4.27	4.90	5.64	14.3%
UK	2.00	2.28	2.60	2.97	3.39	3.89	4.47	5.17	14.7%
France	1.58	1.79	2.03	2.29	2.60	2.96	3.37	3.86	13.7%
Italy	0.45	0.50	0.57	0.64	0.72	0.82	0.93	1.07	13.4%
Spain	0.34	0.39	0.43	0.49	0.55	0.62	0.70	0.79	12.8%
Rest of Europe	3.76	4.21	4.72	5.29	5.93	6.66	7.52	8.52	12.5%
Total	10.37	11.72	13.24	14.97	16.93	19.22	21.89	25.05	13.6%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis